RANDOM VARIABLES

In a Random experiment if a real variable is associated with every outcome, then it is called a random variable or stochastic variable. In other words, a random variable is a function that assigns a real number to every sample point in the sample space. Random variables are usually denoted by capital letters. It may be noted that diff. random variables may be associated with same sample space.

The set of real no.s. of a random variable is called the range of X.

Suppose a coin is tossed twice, we shall associate to diff ~~kind of~~ random variables X and Y as follows:

X = Number of heads in the outcome

Y = Number of tails in the outcome

We have $S = \{HH, HT, TH, TT\}$

The association of elements in S to X is as follows

Outcome	HH	HT	TH	TT
X	2	1	1	0

Illy the association of ele in S to Y

Outcome	HH	HT	TH	TT
Y	0	1	1	2

∴ The range of $X = \{0,1,2\}$ and of $Y = \{0,1,2\}$

$P(X=0) = \frac{1}{4}$ $P(X=1) = \frac{2}{4} = \frac{1}{2}$ $P(X=2) = \frac{1}{4}$

Discrete and Continuous random variables :

If a random variable takes finite or countable infinite number of values, then it is called discrete random variables.

If a random variable takes non-countable infinite no. of values, then it is called continuous random variables.

Eg:- Tossing the coin and observing outcome

Throwing a die and observing no. of on the face

- Observing a phr on a speedometer (continuous random
- conducting a survey on life of electric bulbs.

Discrete probability distribution

If for each value x_i for a discrete random variable X, we assign a real no. $p(x_i)$ such that

i) $p(x_i) \geq 0$

& ii) $\sum_i p(x_i) = 1$

Then the funct^n $p(x)$ is called a probability function.

If the prob that X takes the values x_i is P_i then $P(X = x_i) = P_i$ or $P(x_i)$

The set of values $\{x_i, p(x_i)\}$ is called a discrete probability distribution. The function $p(x)$ is called the probability density function.

Eg: 1) From a sealed box containing a dozen apples, it was found that 3 apples were perished. Obtain the prob distribution of the no. of perished apples when 2 apples are drawn at random.

Soln: Let X be the no. of perished apples.

Since there are 12 apples, 2 apples can be drawn in $^{12}C_2$ ways.

$\therefore X = \{0, 1, 2\}$

Probability of getting 0 perished apples is

$$P(X=0) = \frac{^3C_0 \times {}^9C_2}{^{12}C_2} = \frac{1 \times \frac{9 \times 8^2}{2\cancel{\times 1}}}{\frac{3\cancel{12} \times 11}{2\cancel{\times 1}}} = \frac{\overset{6}{\cancel{18}}}{\cancel{3} \times 11} = \frac{6}{11}$$

Prob. of getting 1 perished apple is

$$P(X=1) = \frac{^3C_1 \times {}^9C_1}{^{12}C_2} = \frac{3 \times 9}{\frac{\overset{6}{\cancel{12}} \times 11}{\cancel{2}}} = \frac{9}{22}$$

Prob. of getting 2 perished apples is

$$P(X=2) = \frac{^3C_2 \times {}^9C_0}{^{12}C_2} = \frac{\frac{3 \times \cancel{2}}{\cancel{2}} \times 1}{\frac{6\cancel{12} \times 11}{\cancel{2}}} = \frac{1}{22}$$

Probability distribution

x	0	1	2
$P(x)$	$\frac{6}{11}$	$\frac{9}{22}$	$\frac{1}{22}$

1) Find the value of 'k' such that the following distribution represents discrete probability distribution. Hence find its mean and std deviation. Also find

i) $P(x<=1)$ ii) $P(x>1)$ iii) $P(-1<x\leq 2)$

x	-3	-2	-1	0	1	2	3
$f(x)$	k	2k	3k	4k	3k	2k	k

since $\{x_i, p(x)\}$ is a discrete (finite) probability distribution.

we must have $p(x_i) \geq 0$ which will be true only if $k \geq 0$.

& $\sum_{i=1}^{7} p(x_i) = 1$

$k + 2k + 3k + 4k + 3k + 2k + k = 1$

$16k = 1$

$\therefore k = 1/16$

Mean $(\mu) = \sum x_i p(x_i)$

$= -3k - 4k - 3k + 0 + 3k + 4k + 3k = 0$

variance $(\sigma^2) = \sum (x_i - \mu)^2 p(x_i)$ $\quad \mu = 0$

$= 9k + 8k + 3k + 0 + 3k + 8k + 9k$

$= 40k = 40 \times \frac{1}{16} = \frac{10}{4} = \frac{5}{2}$

$= 2.5$

Std deviation $(\sigma) = \sqrt{v} = \sqrt{2.5}$

i) $P(x \leq 1) = \sum_{i}^{5} p(x_i) = k + 2k + 3k + 4k + 3k = 13k = \frac{13}{16}$

ii) $p(x>1) = p(x=2) + p(x=3) = 2k + k = \frac{3}{16}$

iii) $p(-1<x\leq 2) = p(x=0) + p(x=1) + p(x=2)$

$= 4k + 3k + 2k = \frac{9}{16}$

2) A random variable X has $p(x) = 2^{-x}$ $x = 1, 2, 3, 4, \ldots$ S.T. $p(x)$ is a probability function. Also find i) $P(X$ is even$)$

ii) $p(x$ being divisible by $3)$ iii) $P(X \geq 5)$

1/4 1/16

Soln: Clearly $p(x) = 2^{-x} = \frac{1}{2^x} \geq 0$ for $x = 1, 2, 3, 4, \ldots$

geometric series. $r = \frac{1}{2}$, $a = \frac{1}{2}$

Q. $\sum_{x=1}^{\infty} p(x) = \frac{1}{2^1} + \frac{1}{2^2} + \frac{1}{2^3} + \frac{1}{2^4} + \cdots$

$= \frac{a}{1-r} = \frac{\frac{1}{2}}{1-\frac{1}{2}} = \underline{\underline{1}}$

$\therefore p(x) = 2^{-x}$ is a probability function.

i) $P(x \text{ is even}) = P(x=2) + P(x=4) + P(x=6) + \cdots$

$= \sum_{x=2,4,6}^{\infty} p(x_i) = \frac{1}{2^2} + \frac{1}{2^4} + \frac{1}{2^6} + \cdots$

This is a geometric series with common ratio, $r = \frac{1}{2^2} = \frac{1}{4} < 1$

$\therefore P(x \text{ is even}) = \frac{a}{1-r} = \frac{\frac{1}{4}}{1-\frac{1}{4}} = \frac{\frac{1}{4}}{\frac{3}{4}} = \underline{\underline{\frac{1}{3}}}$

ii) $P(X \text{ being divisible by } 3) = P(x = 3, 6, 9, 12, \ldots)$

$= \sum_{x=3,6,9}^{\infty} p(x_i) = \frac{1}{2^3} + \frac{1}{2^6} + \frac{1}{2^9} + \cdots$

This is a geometric series with $r = \frac{1}{2^3} = \frac{1}{8} < 1$

$\therefore P(X \text{ divisible by } 3) = \frac{\frac{1}{8}}{1-\frac{1}{8}} = \underline{\underline{\frac{1}{7}}}$

iii) $P(x \geq 5) = P(x = 5, 6, 7, 8, 9, \ldots)$

$= \frac{1}{2^5} + \frac{1}{2^6} + \frac{1}{2^7} + \cdots$

$a = \frac{1}{2^5} = \frac{1}{32}$, $r = \frac{1}{2}$

$\therefore P(x \geq 5) = \frac{\frac{1}{32}}{1-\frac{1}{2}} = \frac{2}{32} = \underline{\underline{\frac{1}{16}}}$

3) A random variable X has the following probability fn for various x.

x	0	1	2	3	4	5	6	7
P(x)	0	k	2k	2k	3k	k^2	$2k^2$	$7k^2+k$

i) Find constant k.

ii) Evaluate $P(x \leq 6)$, $P(x \geq 6)$, $P(3 < x \leq 6)$.

iii)

4) A random expt of throwing a die twice is performed. A random variable X on this sample space is defined to be the sum of 2 ~~variables~~ nos turning up. Find the discrete prob. distribution for the random variable X.

and compute the corresponding mean and std. deviation.

3) Soln:
since X is a discrete random variable, we must have

i) $p(x) \geq 0$, which is true only if $k \geq 0$.

ii) $\sum p(x) = 1$

$$0 + k + 2k + 2k + 3k + k^2 + 2k^2 + 7k^2 + k = 1$$

$$10k^2 + 9k = 1$$

$$k(10k + 9) = 1$$

$k = 1$ and $10k + 9 = 1$

$$k = \frac{-8}{10} = \frac{-4}{5} < 0$$

$$10k^2 + 9k - 1 = 0$$

$$10k^2 + 10k - k - 1 = 0$$

$$10k(k+1) - 1(k+1) = 0$$

$k = -1$ and $10k - 1 = 0$

$$k = \frac{1}{10}$$

$$P(X < 6) = P(X = 0, 1, \ldots 5)$$

$$= 0 + k + 2k + 2k + 3k + k^2$$

$$= k^2 + 8k$$

$$= \frac{1}{100} + \frac{8}{10} = \frac{81}{100}$$

$$P(X \geq 6) = P(X = 6, 7) = 1 - P(X < 6) = 1 - \frac{81}{100} = \frac{19}{100}$$

$$P(3 < x \leq 6) = P(X = 4, 5, 6) = 3k + k^2 + 2k^2 = 3k + 3k^2 = 3k(1 + k)$$

$$= 3\left(\frac{1}{10}\right)\left(1 + \frac{1}{10}\right) = \frac{3}{10}\left(\frac{11}{10}\right) = \frac{33}{100}$$

4) Soln: $S = \{(x, y) : x, y = 1, 2, 3, 4, 5, 6\}$

i.e, $S = \{(1,1), (1,2), (1,3) \ldots (1,6), (2,1) \ldots (6,6)\}$

Total outcomes possible $n(S) = 36$

We have $X = \{2, 3, 4, \ldots 11, 12\}$ (sum)

$X = 2$ $(1,1)$ $P(X=2) = \frac{1}{36}$

$X = 3$ $(1,2), (2,1)$ $P(X=3) = \frac{2}{36}$

$X = 4$ $(1,3), (3,1), (2,2)$ $P(X=4) = \frac{3}{36}$

$X = 5$ $P(X=5) = \frac{4}{36}$

$X = 6$ $P(X=6) = \frac{5}{36}$

$X = 7$ $P(X=7) = \frac{6}{36}$

$8 = \frac{5}{36}$ $9 = \frac{4}{36}$ $10 = \frac{3}{36}$ $11 = \frac{2}{36}$ $12 = \frac{1}{36}$

$X = 12$ $(6,6)$ $P(X=6) = \frac{1}{36}$

Mean(μ) $= \sum x_i p(x_i)$

$$= \frac{140}{36} + \frac{112}{36} = \frac{252}{36} = 7$$

Variance $= 2\left(\frac{105}{36}\right) = 5.833$ $\rightarrow (x_i - \mu)^2 p_i$

Std. dev $= 2.4152$

1) If a random variable X take the values 1, 2, 3, 4 such that

$2P(X=1) = 3P(X=2) = P(X=3) = 5P(X=4)$, find the probability distribution function of X.

2) If X is a discrete random variable taking values 0, 1, 2, 3, etc with $P(X) = \frac{1}{2}\left(\frac{2}{3}\right)^x$, find $P(x$ being an odd no) by first establishing that $P(X)$ is a probability function.

3) X is a discrete random variable having $P(x)$ defined as follows

$$P(x) = \begin{cases} \frac{x}{15} \text{ if } & 1 \le x \le 5 \\ 0 \quad : & x > 5 \end{cases} \quad \text{s.t. } P(x) \text{ is a prob. function. Also find}$$

i) $P(X = 1 \text{ or } 2)$ ii) $P\left(\frac{1}{2} < x < \frac{5}{2} \mid x > 1\right)$.

Soln:

1) Let $\{X, P(X)\}$ be the discrete probability distribution

X	1	2	3	4
$P(X)$	P_1	P_2	P_3	P_4

We have to find P_1, P_2, P_3, P_4 such that their sum is 1,

$$P_1 + P_2 + P_3 + P_4 = 1$$

Given:- $2P_1 = 3P_2 = P_3 = 5P_4$

$2P_1 = 3P_2 \Rightarrow P_2 = \frac{2}{3}P_1$

$2P_1 = 5P_4 \Rightarrow P_4 = \frac{2}{5}P_1$

$$P_1 + \frac{2}{3}P_1 + 2P_1 + \frac{2}{5}P_1 = 1$$

$$\Rightarrow 3P_1 + \frac{16}{15}P_1 = 1 \Rightarrow \frac{45P_1 + 16P_1}{15} = 1 \Rightarrow \frac{61}{15}P_1 = 1$$

$$\therefore \boxed{P_1 = \frac{15}{61}}$$

$$P_2 = \frac{2}{3}P_1 = \frac{2}{3} \times \frac{15}{61} = \frac{10}{61}$$

$$P_3 = 2P_1 = 2 \times \frac{15}{61} = \frac{30}{61}$$

$$P_4 = \frac{2}{5}P_1 = \frac{2}{5} \times \frac{15}{61} = \frac{6}{61}$$

2) $P(x) = \frac{1}{2}\left(\frac{2}{3}\right)^x$ for $x = 1, 2, 3, \ldots$

clearly $P(x) \ge 0$

and consider $\sum_1^\infty P(x) = \frac{1}{2}\left(\frac{2}{3}\right) + \frac{1}{2}\left(\frac{2}{3}\right)^2 + \frac{1}{2}\left(\frac{2}{3}\right)^3 + \ldots$

$$= \frac{1}{2}\left(\frac{2}{3}\right)\left[1 + \frac{2}{3} + \left(\frac{2}{3}\right)^2 + \ldots\right]$$

G.P. $a = 1$, $r = \frac{2}{3} < 1$

$= \frac{1}{2}\left(\frac{2}{3}\right) \cdot \frac{1}{\left(1-\frac{2}{3}\right)}$ $\quad \frac{a}{1-r}$

$= \frac{1}{3} \cdot \frac{3}{1}$

$\therefore \sum P(x) = 1$

$\therefore P(x)$ is probability function

$P(x) = \frac{1}{2}\left(\frac{2}{3}\right)^x$ for $x = 1, 2, 3, \ldots$

$P(x$ being odd number$) = \frac{1}{2}\left(\frac{2}{3}\right)^1 + \frac{1}{2}\left(\frac{2}{3}\right)^3 + \frac{1}{2}\left(\frac{2}{3}\right)^5 + \frac{1}{2}\left(\frac{2}{3}\right)^7 + \ldots$

$= \frac{1}{2}\left(\frac{2}{3}\right)\left[1 + \frac{4}{9} + \left(\frac{4}{9}\right)^2 + \ldots\right]$ G.P. $a = 1$; $r = \frac{4}{9} < 1$

$= \frac{1}{2}\left(\frac{2}{3}\right) \cdot \frac{1}{1-\frac{4}{9}}$

$= \frac{1}{3} \cdot \frac{9}{5}^3$

$= \frac{3}{5} = 0.6$

3) i) $P(x) \geq 0$

ii) Sum, $\sum_1^\infty P(x) = \sum_1^5 P(x) + \sum_5^\infty P(x)$

$= \frac{1}{15} + \frac{2}{15} + \frac{3}{15} + \frac{4}{15} + \frac{5}{15} + 0 + 0 + \ldots$

$= \frac{15}{15} + 0$

$= 1$

$\therefore P(x)$ is a probability function.

i) $P(X=1 \text{ or } 2) = P(X=1) + P(X=2)$

$= \frac{1}{15} + \frac{2}{15} = \frac{3}{15} = \frac{1}{5}$

ii) $P\left(\frac{1}{2} < X < \frac{5}{2} \mid X > 1\right) = \frac{P\left(\frac{1}{2} < X < \frac{5}{2} \cap X > 1\right)}{P(X>1)} = \frac{P\left(\frac{1}{2} < X < \frac{5}{2}\right) \cdot P(X>1)}{P(X>1)}$

$=$

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Binomial distribution / Bernoulli theorem

A random expt with only 2 possible outcomes categorized as success and failure is called a Bernoulli trial, where the prob. of success 'p' is same for each trial.

Bernoulli's theorem :

The prob. of 'x' successes in 'n' trials is ${}^nC_x p^x q^{n-x}$ where,

p → prob. of success

q → prob of failure.

Proof: Since p is the prob. of success, the prob. of x successes is

$p.p.p.p. \ldots$ x times $= p^x$

Also x successes imply $(n-x)$ failures and since q is the prob of failure, the prob of $(n-x)$ failures is $q.q.q \ldots (n-x)$ times $= q^{n-x}$

By the multiplication rule, the prob. of simultaneous happening is $p^x q^{n-x}$.

But x successes in n trials can occur in ${}^nC_x$ ways and all these cases are favourable to the event. Hence by the addition rule, the prob of x successes out of n trails is given by

$p^x q^{n-x} + p^x q^{n-x} + \ldots$ ${}^nC_x$ times $= {}^nC_x p^x q^{n-x}$

The prob. distribution $\{x, p(x)\}$ is given by

x	0	1	2	3 ...	n-1	n
p(x)	q^n	${}^nC_1 p^1 q^{n-1}$	${}^nC_2 p^2 q^{n-2}$	...	${}^nC_{n-1} p^{n-1} q$	p^n

${}^nC_0 p^0 q^{n-0} + {}^nC_1 p^1 q^{n-1} + {}^nC_2 p^2 q^{n-2} + \ldots {}^nC_{n-1} p^{n-1} q^{n-(n-1)} + {}^nC_n p^n q^{n-n}$

$q^n + npq^{n-1} + {}^nC_2 p^2 q^{n-2} + \ldots + {}^nC_{n-1} p^{n-1} q + p^n$

$= (p+q)^n$

$= 1^n$

$= 1$

It may be observed that value of p^x for different values $x = 0, 1, 2, 3, \ldots n$ are the successive terms in the binomial expansn of $(q+p)^n$ and

accordingly this distribn is called the binomial distribution.

- The mean(μ) value of binomial distribution is np.
- The std. deviation of the , $\sigma = \sqrt{npq}$

1) The prob. that a person aged 60 years will live upto 70 years is 0.65. what is the prob. that out of 10 persons aged 60 atleast 7 of them will live upto 70?

$p = 0.65$ $\quad n = 10$ $\quad$ X - no. of persons aged 60 yrs living upto 70 years. $\quad {}^nC_x p^x q^{n-x}$

$q = 0.35$

$P(\text{atleast } 7) = P(X=7) + P(X=8) + P(X=9) + P(X=10)$

$= {}^{10}C_7 (0.65)^7 (0.35)^3 + {}^{10}C_8 (0.65)^8 (0.35)^2 + {}^{10}C_9 (0.65)^9 (0.35)^1 + {}^{10}C_{10} (0.65)^{10} (0.35)^0$

$= \frac{10 \times 9^3 \times 8^4}{3 \times 2} (0.049)(0.042) + \frac{10^5 \times 9}{2} (0.031)(0.1225) + 10 \times (0.02)(0.35) + 1(0.013)$

$= 0.2469 + 0.17 + 0.07 + 0.013$

$= 0.499$

$P(X \geq 7) = 0.5138$